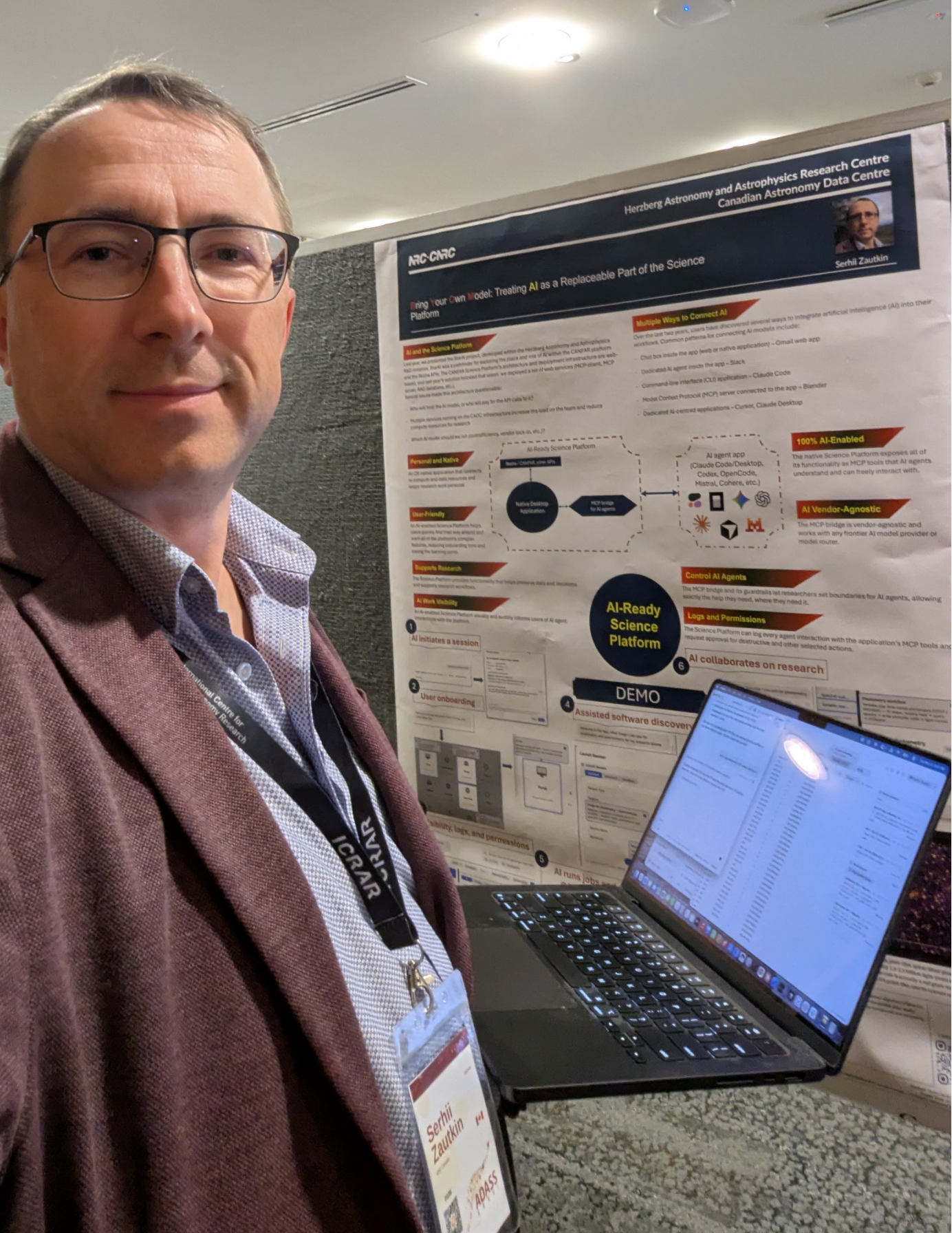

NRC·CNRC
Herzberg Astronomy and Astrophysics Research Centre
Canadian Astronomy Data Centre
Serhil Zautkin
Bring Your Own Model: Treating AI as a Replaceable Part of the Science Platform
AI and the Science Platform
Multiple Ways to Connect AI
Over the last two years, users have discovered several ways to integrate artificial intelligence (AI) into their workflows. Common patterns for connecting AI models include:
Chat box inside the app (web or native application) – Gmail web app
Dedicated AI agent inside the app – Slack
Command-line interface (CLI) application – Claude Code
Model Context Protocol (MCP) server connected to the app – Blender
Dedicated AI-centred applications – Cursor, Claude Desktop
Personal and Native
User-Friendly
Supports Research
AI Work Visibility
AI-Ready Science Platform
Native / Colorful, crisp, APIs...
Native Desktop Application
MCP bridge for AI agents
AI agent app (Claude Code/Desktop, Codex, OpenCode, Mistral, Cohere, etc.)
100% AI-Enabled
The native Science Platform exposes all of its functionality as MCP tools that AI agents understand and can freely interact with.
AI Vendor-Agnostic
The MCP bridge is vendor-agnostic and works with any frontier AI model provider or model router.
AI-Ready Science Platform
Control AI Agents
The MCP bridge and its guardrails let researchers set boundaries for AI agents, allowing exactly the help they need, where they need it.
Logs and Permissions
The Science Platform can log every agent interaction with the application's MCP tools and request approval for destructive and other selected actions.
1 AI initiates a session
2 User onboarding
3 visibility, logs, and permissions
4 Assisted software discovery
5 AI runs jobs
6 AI collaborates on research
DEMO
ICRAR
Serhii Zautkin
ADASS